

FINANCIAL MATH & ANALYSIS CONCEPTS (190)

—OPEN EVENT—

STATE – 2019

DO NOT WRITE ON TEST BOOKLET

TOTAL POINTS _____ (*100 points*)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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GENERAL INSTRUCTIONS

The number of decimal places used in calculations and in the reporting of your answers is very important. Below are guidelines to use as you take this test. If a problem varies from these guidelines, it will be noted in the problem.

PAYROLL QUESTIONS

- Do not round the overtime rate.
- Overtime is paid for any hours over 40 worked in a week, *unless otherwise stated in the problem.*
- Normal work week is 40 hours, *unless otherwise stated in the problem.*

PERCENTAGE PROBLEMS

- Round to the nearest tenth when working with percentages.

Example: Use 2.3% NOT 2%

Example: An answer of .9638 should be written as 96.4%

FRACTIONS

- All problems with fractions are to be reduced to their lowest terms.

INTEREST

- All interest rates are stated in one-year simple interest, *unless otherwise noted.*
- Interest is to be calculated on 365 days.

DOLLAR AMOUNTS

- All dollar amounts must be rounded to the second place (nearest cent).

Example: \$35.9765 to \$35.98

MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question.

1. In her summer job, McKenzie Lopez is paid \$300 a week and a commission of 10% on sales above \$3,500. Her sales last week were \$3,904. What were her total earnings for the week?
 - a. \$340.40
 - b. \$294.25
 - c. \$690.40
 - d. \$3,923.25

2. Corey paid \$12,400 for his car. Annual payments for insurance, gas, oil, repairs and other expenses total \$1,849. The car depreciates 11% a year. What was his total annual cost of operating the car?
 - a. \$1,364
 - b. \$485
 - c. \$3,213
 - d. \$3,509

3. Danni Cuoio was injured on her job. She is insured by a disability policy that pays 65% of her average annual salary of \$66,800. What amount does Danni collect monthly?
 - a. \$434.20
 - b. \$3,618.33
 - c. \$36,183.33
 - d. \$43,420.00

4. The regular price of a frozen drink at the local coffee shop is \$3.75. You can buy a 12-punch, punch card for \$25.00. How much will you save by buying the pre-paid punch card?
 - a. \$16.80
 - b. \$20.00
 - c. \$12.50
 - d. It is cheaper to buy the drinks individually.

5. West Credit Union charges a \$35 per check or debit card transaction overdraft protection fee. On January 9, Ritchie had \$385 in his account. Over the next four days, the following debit card transactions or checks arrived for payment: January 10, \$356; January 11, \$159 and \$120; January 12, \$19.87; January 13, \$5.66. Using the information above, how much will Ritchie pay in overdraft protection fees?
 - a. \$70
 - b. \$105
 - c. \$140
 - d. \$175

6. What is another name for simple average?
- estimate
 - mean
 - best approximation
 - median
7. When comparing credit card offers, all of the following should be evaluated before making a decision with the *exception* of ____.
- the annual percentage rate
 - the credit limit available
 - the method used for calculating finance charges
 - the annual fees assessed
8. The Professional Business Associates purchases wholesale commemorative teddy bears for its 50th Anniversary at \$23.00. If the online store marks each bear up 110%, what is the price of the bear?
- \$25.30
 - \$27.30
 - \$50.60
 - \$48.30
9. Jennifer bought 200 \$600, 9.25% bonds. What is her annual income from the bonds?
- \$1,200
 - \$11,100
 - \$1,100
 - \$110
10. John Thompson embroiders baseball caps. He is paid \$2.50 for each cap he embroiders. Thompson's production sheet is shown below. What is his total pay for the week?

DATE	HATS
6/14	42
6/15	37
6/16	0
6/17	24
6/18	41

- \$375.00
- \$350.00
- \$360.00
- \$345.00

11. A cupcake shop has current assets of \$60,100 and current liabilities of \$32,100. Which of the following best represents the shop's current ratio?

- a. 28,000
- b. 2:1
- c. 1.9:1
- d. 0.53:1

12. Latoya Williams' savings account statement is shown below. What is her new balance?

Previous Balance	+	Interest	+	Deposits	-	Withdrawals	=	New Balance
\$968.75		\$8.37		\$27.00		\$138.93		

- a. \$866.91
- b. \$865.19
- c. \$866.19
- d. \$1,143.05

13. Frank Peterson recently opened The Sporting Goods Store. He used \$80,000 of his own money and took out a bank loan of \$35,000. From the \$115,000 cash, Frank paid \$20,000 for merchandise and \$20,000 for supplies. This left a cash balance of \$75,000. He received another shipment of \$6,000 worth of merchandise. He did not pay for this merchandise immediately. What was his owner's equity?

- a. \$ 80,000
- b. \$120,000
- c. \$ 41,000
- d. \$100,000

14. The Moy family budgeted \$32,650 to remodel their kitchen. They spent \$43,987. How much more or less is the amount spent than the amount budgeted?

- a. \$11,337 more
- b. \$11,337 less
- c. \$17,887 more
- d. \$17,887 less

15. Jason Cooper's gross weekly pay is \$598. His earnings to date for the year total \$14,950. What total amount is deducted from his pay this week for Social Security, which is taxed at 6.2 percent, and Medicare, at 1.45 percent?

- a. \$39.68
- b. \$45.74
- c. \$40.39
- d. \$45.75

16. Linda Griffith is planning to buy rental property. She finds a home priced at \$92,000. What is the amount of Griffith's mortgage loan if a down payment of 10 percent is required?
- \$81,800
 - \$92,610
 - \$87,900
 - \$82,800
17. Dan Miller has been granted a \$120,000 mortgage loan. When signing the papers to purchase his new home, he will need to pay the closing costs shown below. What are the total closing costs?

Credit Report	\$ 65
Loan Origination Fee: $\$120,000 \times 2.0\%$	2,400
Abstract of Title	120
Attorney Fee	300
Documentation Stamp: $\$120,000 \times 0.3\%$	360
Processing Fee: $\$120,000 \times 1.10\%$	1,320

- \$4,500
 - \$3,270
 - \$4,565
 - \$4,650
18. Sharon Hensley and her brother purchased a house for \$156,600. Neighbors Credit Union financed their loan for 30 years at 7% APR with 20 percent down. For the first monthly payment, what is the interest portion of the payment?
- \$730.80
 - \$763.30
 - \$856.60
 - \$942.30
19. Cristina Suarez has insured her home for 80 percent of its replacement value of \$155,000. Use the chart below to determine the amount of coverage for her personal property.

Insurance Coverage Chart	
Coverage	Percent of Coverage
Garage and Other Structures	20%
Loss of Use	25%
Personal Property	45%

- \$55,800
- \$56,900
- \$44,230
- \$51,092

20. Stanley McBride is employed by the True Blue Company. The PPO annual premium is \$7,058. His employer pays 70 percent of the total cost. His contribution is deducted from his paycheck. What is McBride's monthly deduction?
 - a. \$200.14
 - b. \$176.45
 - c. \$56.95
 - d. \$152.87
21. Thomas Trotte is 50 years old. He wants to purchase a \$50,000, 5-year term life insurance policy. The premium per \$1,000 is \$5.80. What is his annual premium?
 - a. \$560.00
 - b. \$296.00
 - c. \$290.00
 - d. \$580.00
22. Keenan Williams took out a \$100,000, 5-year term policy at age 40. The premium per \$1,000 was \$3.56. She will be 45 years old this year. The premium per \$1,000 will be \$4.62. What is the percent increase?
 - a. 36%
 - b. 30%
 - c. 42%
 - d. 32%
23. Florence Tyler invests \$6,500 in a 4-year certificate of deposit that earns interest at an annual rate of 5 percent compounded daily. The FV factor per \$1.00 is 1.221386. What is the interest earned?
 - a. \$23,906.00
 - b. \$1,439.01
 - c. \$165.28
 - d. \$1,896.50
24. Brian Gray buys 1,200 shares of food company stock at \$38.20 per share. His online broker charged him \$0.03 per share for commission. What is the total cost of the stock?
 - a. \$45,876.00
 - b. \$45,982.01
 - c. \$58,210.98
 - d. \$60,129.34
25. Maya Franklin purchased a \$3,500 bond at the quoted price of $80 \frac{1}{2}$. The bond paid interest at a rate of 5 percent. What is the cost of the bond?
 - a. \$8,251.30
 - b. \$5,656.11
 - c. \$2,817.50
 - d. \$4,589.14

26. Alanzo's monthly gross pay is \$6,500.00. The deductions from his monthly pay include \$840.00 for federal income tax, \$487.96 for FICA taxes, \$250.00 for his retirement plan, and \$450.00 for health insurance. Find the percentage of his gross wages that he takes home to the nearest percent.
- 35%
 - 45%
 - 69%
 - 220%
27. If a square has a perimeter of 760 feet, what is the area of the square?
- 380 square feet
 - 760 square feet
 - 32,100 square feet
 - 36,100 square feet
28. You have a credit card balance of \$1,300. Payments made to the company are applied to interest first and then to balance. The interest is calculated on the balance you owe. Your credit card carries an annual rate of 22.5%. If you make one monthly \$25 payment, how much will go toward the balance owed after the interest is deducted from the payment?
- \$0.48
 - \$0.56
 - \$0.63
 - \$0.65
29. Fraction $\frac{1}{7}$ is to $\frac{5}{35}$ as $\frac{1}{2}$ is to ____/30.
- 12
 - 15
 - 18
 - No correlation exists
30. Bob's business bought a new copier for \$7,800. The resale value is estimated at \$1,400 after 5 years of use. What is the annual depreciation using the straight-line method?
- \$1,280
 - \$1,560
 - \$280
 - \$1,300
31. An invoice for \$612 has cash discount terms of 2/10, 3/15, n/30 and is dated March 24. How much should be paid on May 1?
- \$599.76
 - \$593.64
 - \$459.00
 - \$612.00

32. Smith, Jones, and Murphy have invested the following into their partnership: Smith = \$25,000, Jones = \$40,000, and Murphy = \$35,000. The partnership agreement dictates that all profits or losses are to be shared in the ratio of partners' investments. Last year the partnership realized a profit of \$38,425. What is Murphy's share?
 - a. \$9,606.25
 - b. \$15,370.00
 - c. \$13,448.75
 - d. \$12,808.33
33. The Butler County tax assessor determines that the market value of Greg Wilson's home is \$123,000. The rate of assessment in Butler County is 28 percent of market value. What is the assessed value of Wilson's home?
 - a. \$34,440
 - b. \$32,220
 - c. \$32,550
 - d. \$32,660
34. Claire Fisher took out a \$60,000, 5-year term policy at age 44. The premium per \$1,000 was \$4.35. She will be 49 years old this year. The premium per \$1,000 will be \$6.22. What is the percent increase?
 - a. 78%
 - b. 42%
 - c. 56%
 - d. 43%
35. Tonya Cox is an assistant to the city manager. Her annual salary is \$45,000. Next month she will receive a yearly cost-of-living adjustment of 2.5 percent. In addition, her manager plans to award her a merit increase of 4 percent for her excellent work during the year. What will Cox's salary be for the coming year?
 - a. \$45,458
 - b. \$47,925
 - c. \$45,487
 - d. \$79,314
36. An operator can make 40 hinges per hour. The direct labor cost is \$17.50 per hour. What is the direct labor cost per hinge?
 - a. \$0.438
 - b. \$0.569
 - c. \$2.689
 - d. \$1.589

37. Dan Miller works as a customer service representative for an answering service. He works 8 hours a day. He spends 7.75 hours each day answering telephone calls and 15 minutes per day doing miscellaneous activities. What percentage of Miller's day is spent answering the telephone?
- 96.88%
 - 85.45%
 - 75.50%
 - 64.80%
38. The video Store sells used DVDs for \$11.98. The Video Store's cost for these DVDs is \$6.83. What is the markup rate based on the selling price?
- 57.0%
 - 75.4%
 - 35.2%
 - 43.0%
39. Out of 2,700 office workers in a marketing research study, 953 preferred a new computer printer ink cartridge. The estimated market size is 1,700,000. An office worker uses about 42 ink cartridges each year. What is the annual sales potential rounded to the nearest hundred thousand?
- 71,400,000 cartridges
 - 35,700,000 cartridges
 - 28,300,000 cartridges
 - 25,200,000 cartridges
40. While visiting, Barbara Galarza's aunt trips on the stairs and twists her ankle. She decides to hold Galarza liable for the accident and expects her to cover her medical expenses. Mutual Farm Insurance provides coverage on the home at 90 percent of its \$275,000 replacement value. Use the chart below to determine the amount of medical coverage.

Mutual Farm Insurance Coverage Chart	
Coverage	Percent Of Coverage
Garage And Other Structures	20%
Personal Property	50%
Loss Of Use	15%
Personal Liability	50%
Medical Payments	3%

- \$6,742
- \$7,425
- \$6,891
- \$7,642

41. Simon Kovach can invest \$8,000 in a 1-year CD that earns interest at an annual rate of 4 percent compounded monthly. The amount per \$1.00 is 1.040742. He can also invest \$7,000 in a 1-year CD at an annual rate of 4 percent compounded quarterly. The amount per \$1.00 is 1.040604. What is the difference in the amount of interest earned for each investment?
 - a. \$21.99
 - b. \$41.71
 - c. \$45.99
 - d. \$23.09
42. Johnson Flooring hired a new national distribution manager for an annual salary of \$178,450. Advertising costs were \$1,341. Interview costs totaled \$713.98. The recruiter fee was 15 percent of the first year's salary. What was the total cost of hiring the distribution manager?
 - a. \$32,987.45
 - b. \$28,124.00
 - c. \$45,598.00
 - d. \$28,822.48
43. The selling price of a product is \$0.74 per unit. The total fixed cost for manufacturing is \$20,456 and variable costs total \$0.34 per unit. How many units must be sold for the manufacturer to break even?
 - a. 62,789
 - b. 51,140
 - c. 50,149
 - d. 52,459
44. Desi Munoz tests motherboards for a computer manufacturer. He works 48 minutes every hour for 8 hours a day. He spends 20 minutes testing each unit. How many motherboards should Munoz be able to test in one work day?
 - a. 15.5
 - b. 19.2
 - c. 14.5
 - d. 13.5
45. The Traders Inc. catalog lists a vintage poster for \$179.89. The net price for dealers is listed at \$137.50. What is the trade-discount rate?
 - a. 37.54%
 - b. 50.68%
 - c. 25.50%
 - d. 30.83%

46. Graham's Candy Shop had net sales of \$541,290, gross profit of \$162,300 and net income of \$42,460. What was Graham's net profit margin, to the nearest tenth percent?
- a. 7.8%
 - b. 12.7%
 - c. 26.2%
 - d. 30.0%
47. Five percent of the 620 survey returns to a mail survey were *not* valid. If 3,400 surveys were mailed, what was the response rate, to the nearest percent?
- a. 5%
 - b. 9%
 - c. 17%
 - d. 18%
48. Which term describes the amount of money on which a person must pay taxes?
- a. withholding
 - b. net income
 - c. taxable income
 - d. pension
49. Company profits are distributed to shareholders in the form of _____.
- a. dividends
 - b. losses
 - c. par values
 - d. net asset values
50. Which term describes an employee who receives a fixed amount each day for meals, expenses, etc. while on company business?
- a. tip
 - b. per diem
 - c. hourly
 - d. commission